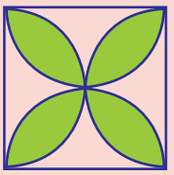
2. If the biggest circle is cut from a square of side '*a*' *units*, then the remaining area in the square is approximately $\frac{3}{14}a^2$ *sq.units.* $\left(\pi = \frac{22}{7}\right)$

3. The area of the biggest circle cut out from the square of 'a' units = $\frac{11}{14}a^2$ *sq. units* (approximately)

4. In the given figure if $\pi = \frac{22}{7}$, the area of the unshaded part of a square of side *a* units is approximately $\frac{3}{7}a^2$ *sq.units* and that of the shaded part is approximately $\frac{4}{7}a^2$ *sq.units.*



Example 2.11

Find the area of an irregular polygon field whose measures are as given in the Fig. 2.30.

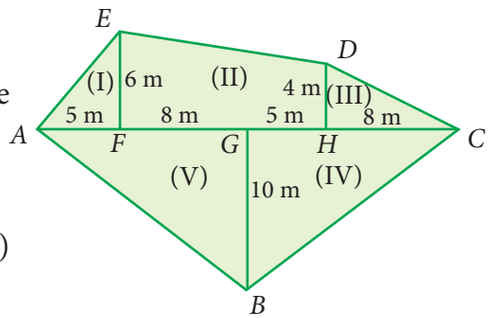


Fig. 2.30

Solution:

The given field has four triangles (I, III, IV and V) and a trapezium (II).

Area of the triangle (I) $= \frac{1}{2} \times b \times h = \frac{1}{2} \times 5 \times 6 = 15 \ m^2$

Area of the trapezium (II) $= \frac{1}{2} h(a+b) = \frac{1}{2} \times 13 \times (6+4) = 65 \ m^2$

Area of the triangle (III) $= \frac{1}{2} \times b \times h = \frac{1}{2} \times 8 \times 4 = \frac{32}{2} = 16 \ m^2$

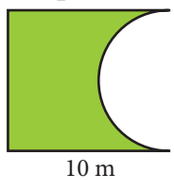
Area of the triangle (IV) $= \frac{1}{2} \times b \times h = \frac{1}{2} \times 13 \times 10 = 65 \ m^2$

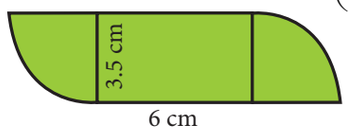
Area of the triangle (V) $= \frac{1}{2} \times b \times h = \frac{1}{2} \times 13 \times 10 = 65 \ m^2$

$\therefore$ The total area of the field $= 15 + 65 + 16 + 65 + 65 = 226 \ m^2$

Exercise 2.2

1. Find the perimeter and area of the figures given below. $\left(\pi = \frac{22}{7}\right)$

(i) 

(ii) 

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